RAKESH CHOWDARY POLINA

rpolina@okstate.edu | https://www.linkedin.com/in/rakesh-chowdary-polina | 405-894-3989

EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Python, Java, JavaScript, SQL, C, C++, HTML, CSS, Swift
Frameworks:	Flask, Django, React JS, Angular JS, Node JS, Spring boot, FastAPI, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Django ORM, SQLAlchemy, OAuth 2, jQuery, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Pip, Poetry

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered and implemented a robust facial recognition pipeline, performing both quantitative and qualitative analysis on image data with Python and InsightFace, to achieve a precision rate of 93.6%
- Implemented DBSCAN clustering and rigorously verified data accuracy across 182,000+ images, enhancing archive reliability and insight precision.
- Designed an intuitive UI for the labeling tool using Flask and React.js, streamlining user engagement and annotations.
- Designed and maintained database structures using SQL, enhancing user interaction management and feedback data integrity, ensuring operational excellence.

Software Engineer, Ericsson, India **Jul 2019 – Jul 2022**

- Transitioned monolithic legacy systems (Java/Spring Boot) to modular microservices using Python and Django REST Framework (DRF), designing RESTful APIs for new features. Integrated Apache Kafka for asynchronous event-driven communication between services, improving scalability to handle 50% more concurrent users.
- Redesigned user interfaces for internal dashboards and customer portals using Angular and TypeScript, implementing reusable components and optimizing rendering performance by 25%. Collaborated with UX teams to ensure alignment with enterprise design standards.
- Enhanced legacy Java/Spring Boot applications by refactoring PostgreSQL data access layers, optimizing complex queries, and introducing connection pooling. Reduced system downtime by 30% through rigorous security patches and Django-based rate limiting for public APIs.
- Developed scalable APIs in Django, reducing average response time by 40% by implementing caching with Redis and optimizing database indexing. Integrated Apache Solr for search-heavy modules, improving query performance by 35% for telecom inventory management systems.
- Containerized applications using Docker and orchestrated deployments on Kubernetes clusters, reducing deployment time by 50%. Automated rollback strategies and health checks to ensure resilience in production environments.
- Streamlined CI/CD pipelines using GitLab CI, integrating automated testing (unit/integration) and security scans. Cut time-to-market by 40% and reduced production incidents by enforcing peer-reviewed merge requests and pipeline-driven Docker builds.
- Modernized legacy Java/Spring Boot services by decoupling monolithic modules, implementing API gateways, and migrating non-critical logic to Python-based microservices. Maintained backward compatibility for enterprise clients during transitions.
- Enforced security best practices across Django and Spring Boot services, including OAuth2 authentication, PostgreSQL column-level encryption, and Kubernetes network policies. Passed internal audits with zero critical vulnerabilities.
- Collaborated in AGILE (SCRUM) teams, delivering features in bi-weekly sprints. Led sprint retrospectives to refine processes and address technical debt in legacy Java systems.
- Documented API specifications (OpenAPI), deployment playbooks, and post-incident reports. Monitored system performance using Grafana dashboards, resolving bottlenecks in Kafka streams and Solr indexing.

PROJECTS

Sleep data analysis - Applied LSTM network and Granger causality techniques to investigate sleep data on a dataset comprising over 5,000 individuals' sleep patterns, unveiling insights into temporal dynamics and causal connections within the dataset

HireOSUgrads App - Developed an iOS application for university students and recruiters, featuring intuitive swipe gestures for seamless interactions. Utilized Swift, followed MVC architecture and integrated SQLite for robust data management

Journal App - Developed an android application, that allows users to store their daily events and memories using React Native, Express JS, Redux, Redis, MongoDB, REST APIs, Git.